

# Evolution of the SCS Runoff Curve Number Method and Its Application to Continuous Runoff Simulation

J. R. Williams<sup>1</sup>; N. Kannan<sup>2</sup>; X. Wang<sup>3</sup>; C. Santhi<sup>4</sup>; and J. G. Arnold<sup>5</sup>

**Abstract:** The Natural Resources Conservation Service (NRCS) [previously Soil Conservation Service (SCS)] developed the SCS runoff curve-number (CN) method for estimating direct runoff from storm rainfall. The NRCS uses the CN method for designing structures and for evaluating their effectiveness. Structural design is usually based on a single event of a certain probability of occurrence. During the years when many floodwater-retarding watershed projects were planned and constructed (1950–1980), the CN equation was used in a continuous mode to evaluate the projects. To operate CN in a continuous mode, runoff was estimated from a daily rainfall record of approximately 30 years. For each day of recorded rainfall, the five-day antecedent rainfall was used to assign a CN<sub>1</sub> (dry condition), CN<sub>2</sub> (average condition), or CN<sub>3</sub> (wet condition), and runoff was estimated with the appropriate CN. With the development of continuous hydrologic simulation models, CN was related directly to soil water content or estimated using rainfall and potential evapotranspiration (PET) to drive an index. Several methods were attempted and used with different degrees of success over a period of many years. The purpose of this study is to describe the evolution of the continuous CN method and its recent developments. Test results on the basis of the direct-link soil-moisture approach and the revised soil-moisture index method are presented for demonstration purposes. The results indicate that the revised soil-moisture index method is robust and produces realistic runoff estimates over a wide range of soil properties. DOI: [10.1061/\(ASCE\)HE.1943-5584.0000529](https://doi.org/10.1061/(ASCE)HE.1943-5584.0000529). © 2012 American Society of Civil Engineers.

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## Introduction

In 1933 the Soil Erosion Service (SES) [later changed to the Soil Conservation Service (SCS)] was established and assigned the task of setting up demonstration conservation projects and overseeing the design and construction of soil conservation measures. At that time, SCS realized the need to obtain hydrologic data and to establish a simple procedure for estimating rates of runoff. With the passage of the Flood Control Act of 1936 (Public Law 74–738), the Department of Agriculture was charged with installing structures for controlling surface runoff and soil erosion. As a first

step, they decided to obtain infiltration rates at many locations throughout the country (Woodward et al. 2002)

To meet the requirements of infiltration data, SCS and other agencies made thousands of infiltrometer runs during the 1930s and 1940s at the experimental sites. To adequately define the hydrologic characteristics of a watershed, decisions were made regarding the number and location of infiltrometers. In the early 1940s, there were some methods for estimating runoff, all of which were based on infiltration data (Woodward et al. 2002).

Sherman (1949) proposed plotting direct runoff versus storm runoff. Mockus (1949), building on this idea, suggested that surface runoff could be estimated from a combination of factors: soil type, land use, areal extent and location, antecedent rainfall, duration and depth of a storm, average annual temperature, and date of storm. Andrews (unpublished data, 1954) grouped infiltrometer data from Texas, Oklahoma, Arkansas, and Louisiana and found that soil-texture class was the only consistent characteristic within each group. He developed a graphical procedure for estimating direct runoff for a combination (soil cover complex) of soil texture, type and amount of cover, and conservation practices (Woodward et al. 2002).

Musgrave (1955) described a hydrologic classification of soils on the basis of their infiltration rate. He grouped all the soils into four basic groups depending on the minimum infiltration capacity. They were A, B, C, and D, with sands in group A and clays in group D. This hydrologic classification system is a major stepping stone for the runoff curve-number (CN) method (Woodward et al. 2002).

The SCS needed a uniform procedure to estimate runoff throughout the country using available data without the need for calibration. The work carried out by Andrews and Mockus on runoff estimation was suitable for ungauged watersheds. Therefore, their work formed the basis for the generalized SCS runoff equation. Mockus modified the generalized approach by hypothesizing the following relationship (Woodward et al. 2002):

<sup>1</sup>Senior Research Scientist, Texas AgriLife Research, Blackland Research and Extension Center, 720 East Blackland Rd., Temple, TX 76502. E-mail: [jwilliams@brc.tamus.edu](mailto:jwilliams@brc.tamus.edu)

<sup>2</sup>Assistant Research Scientist, Texas AgriLife Research, Blackland Research and Extension Center, 720 East Blackland Rd., Temple, TX 76502 (corresponding author). E-mail: [nkannan@brc.tamus.edu](mailto:nkannan@brc.tamus.edu)

<sup>3</sup>Research Assistant Professor, Texas AgriLife Research, Blackland Research and Extension Center, 720 East Blackland Rd., Temple, TX 76502. E-mail: [swang@brc.tamus.edu](mailto:swang@brc.tamus.edu)

<sup>4</sup>Associate Research Scientist, Texas AgriLife Research, Blackland Research and Extension Center, 720 East Blackland Rd., Temple, TX 76502. E-mail: [csanthi@brc.tamus.edu](mailto:csanthi@brc.tamus.edu)

<sup>5</sup>Supervisory Agricultural Engineer, Agricultural Research Service, United States Dept. of Agriculture, Blackland Research and Extension Center, 802 East Blackland Rd., Temple, TX 76502. E-mail: [Jeff.Arnold@ars.udsda.gov](mailto:Jeff.Arnold@ars.udsda.gov)

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$$F/S = Q/P$$

where  $F$  = actual rainfall retention during a storm;  $S$  = potential maximum retention at the start of the storm;  $Q$  = direct runoff; and  $P$  = total rainfall or potential maximum runoff.

The ratios are true as limits when  $P \rightarrow 0$ ,  $Q/P = F/S$ , and as  $P \rightarrow \infty$ ,  $Q/P \approx F/S \rightarrow 1$ .

The equation, after inclusion of initial abstraction and some rearrangements, changes to Eq. 1 given subsequently.

## The SCS CN Method

The basic CN equation (USDA Soil Conservation Service 1972) is

$$Q = \frac{(R - 0.2s)^2}{R + 0.8s}, \quad R > 0.2s \quad Q = 0.0, \quad R \leq 0.2s \quad (1)$$

where  $Q$  = daily runoff;  $R$  = daily rainfall; and  $s$  is a retention parameter. The retention parameter,  $s$ , varies (1) among watersheds, because soils, land use, management, and slope all vary; and (2) with time, because of changes in soil water content. The parameter  $s$  is related to the CN by the SCS equation (USDA Soil Conservation Service 1972)

$$s = 254 \left( \frac{100}{\text{CN}} - 1 \right) \quad (2)$$

The constant in Eq. (2), 254, gives  $s$  in millimeters. Thus,  $R$  and  $Q$  are also expressed in millimeters.  $\text{CN}_2$  (the curve number for moisture condition 2, or average curve number) can be obtained easily for any area by using the SCS hydrology handbook (USDA Soil Conservation Service 1972). The handbook tables consider soils, land use, and management. Assuming that the handbook  $\text{CN}_2$  value is appropriate for a 5% slope, Williams and Izaurralde (2005a) developed the following equation for adjusting that value for other slopes:

$$s_{2s} = s_2 \left( 1.1 - \frac{\text{STP}}{\text{STP} + \exp(3.7 + 0.02117\text{STP})} \right) \quad (3)$$

where  $s_{2s}$  = retention parameter associated with  $\text{CN}_2$  adjusted for slope; and STP = average slope of the watershed (%). The values of  $\text{CN}_2$  and  $\text{CN}_{2s}$  are calculated by substituting  $s_2$  and  $s_{2s}$  into Eq. 2. Values of  $\text{CN}_1$ , the curve number for moisture condition 1 (dry), and  $\text{CN}_3$  (wet) corresponding to  $\text{CN}_2$  are also tabulated in the handbook. For computing purposes,  $\text{CN}_1$  and  $\text{CN}_3$  were related to  $\text{CN}_2$  with the equations

$$\text{CN}_1 = \text{CN}_{2s} - \frac{20(100 - \text{CN}_{2s})}{100 - \text{CN}_{2s} + \exp[2.533 - 0.0636(100 - \text{CN}_{2s})]} \quad (4)$$

$$\text{CN}_3 = \text{CN}_{2s} \exp[(0.00673 \times (100 - \text{CN}_{2s}))] \quad (5)$$

The purpose of this article is to

1. Describe the evolution of the continuous CN method and its recent developments;
2. Compare the direct-link soil-moisture approach and the revised soil-moisture index method at different sites with varying soils in the Bosque Watershed for demonstration purposes;
3. Demonstrate the application of the revised soil-moisture index (SMI) method in a large watershed (Bosque River Watershed) for flow calibration; and
4. Demonstrate the application of the revised SMI method for a river-basin scale study.

## Continuous CN Development

The SCS, now the Natural Resource Conservation Service (NRCS), uses the CN method for designing structures and for evaluating their effectiveness. Structural design is usually based on a single event of a certain probability of occurrence. During the years when many floodwater retarding watershed projects were planned and constructed (1950–1980), the CN equation was used in a continuous mode to evaluate the projects. To operate CN in a continuous mode, runoff was estimated from a daily rainfall record of approximately 30 years. Each day of recorded rainfall was used for runoff simulation. The five-day antecedent rainfall was used to assign a  $\text{CN}_1$  (dry condition),  $\text{CN}_2$  (average condition), or  $\text{CN}_3$  (wet condition), and runoff was estimated with the appropriate CN. The continuous-CN approach was modified (Williams and LaSeur 1976) by replacing the five-day antecedent rainfall with a SMI as part of a daily water-yield model. The revised SMI approach formed the basis for the use of the continuous CN method in several watershed simulation models.

### The Soil-Moisture Index Methods

The SCS continuous CN approach was modified (Williams and LaSeur 1976) by replacing the five-day antecedent rainfall with a SMI as part of a daily water-yield model. The revised SMI was driven by rainfall, runoff, and lake evaporation in the equation

$$\text{SMI}_t = \frac{\text{SMI}_o + \text{RFV}_o}{1.0 + b(\text{SMI}_o + \text{RFV}_o)\text{LE}_o} - Q_o \quad (6)$$

where SMI = soil-moisture index; RFV = rainfall value; LE = lake evaporation (mm);  $b$  = soil-moisture depletion coefficient; and subscripts  $t$  and  $o$  refer to the present and previous day, respectively. The SCS retention parameter  $s$  was related to the revised SMI in the equation

$$s = V - \text{SMI} \quad (7)$$

where  $V$  = maximum value of SMI. When  $V$  is set to 20, CN ranges from 33 to 100.

Hawkins (1978) showed that the maximum difference between RFV and  $Q$  is runoff.

$$V = 1.2s_1 \quad (8)$$

where  $s_1$  is the SCS retention parameter for soil-moisture condition 1 (dry). However, Hawkins also showed that  $s$  can be computed directly if evapotranspiration (ET) is known, using the equation

$$s_t = s_o + \frac{\text{ET}_o - \text{RFV}_o + Q_o}{1.2} \quad (9)$$

where ET = actual evapotranspiration (mm) and subscripts  $t$  and  $o$  refer to the present and previous day, respectively.

### The Direct Soil-Moisture Methods

A trend in the development of continuous simulation models during the 1970s and 1980s linked the retention parameter  $s$  directly to the soil water content. The Chemicals, Runoff, and Erosion from Agricultural Management Systems (CREAMS) (Knisel 1980) model related  $s$  directly to soil water content using the linear equation

$$s = s_1(1 - \text{FFC}) \quad (10)$$

where FFC = fraction of field capacity in the root zone. FFC is computed with the equation

$$FFC = \frac{SW - WP}{FC - WP} \quad (11)$$

where SW = soil water content in the root zone; WP = wilting-point water content (1,500 kPa for many soils) and FC is the field-capacity water content (33 kPa for many soils). The linear Eq. 10 proved inadequate in many soils because it ranges from  $s_1$  at wilting point to 0.0 at field capacity, and  $s = 0.0$  gives CN = 100, which is much too high for many soils, especially sands. Williams et al. (1984) introduced the nonlinear  $s$ -soil water equation

$$s = s_1 \left( 1 - \frac{FFC}{FFC + \exp[w_1 - w_2(FFC)]} \right) \quad (12)$$

Values for  $w_1$  and  $w_2$  are obtained from a simultaneous solution of Eq. (12) according to the assumptions that  $s = s_2$  when  $FCC = 0.6$  and  $s = s_3$ , when  $(ST - FC)/(PO - FC) = 0.5$  with  $s$  is the soil water storage.

$$w_1 = \ln \frac{60}{1 - s_2/s_1} - 60 + 60.w_2 \quad (13)$$

$$w_2 = \frac{\ln\left(\frac{60}{1 - s_2/s_1} - 60\right) - \ln\left(\frac{POFC}{1 - s_3/s_1} - POFC\right)}{POFC - 60} \quad (14)$$

$$POFC = 100 + 50 \cdot \left( \frac{\sum_{l=1}^M (PO_l - FC_l)}{\sum_{l=1}^M (FC_l - WP_l)} \right) \quad (15)$$

where  $s_3$  = CN<sub>3</sub> retention parameter; PO = porosity of soil layer  $l$ ; and  $M$  = number of soil layers. The values of  $s_1$ ,  $s_2$ , and  $s_3$  are obtained from Eqs. (2), (4), and (5). Eqs. (12)–(15) assure that CN<sub>1</sub> corresponds with the wilting point and that the curve number cannot exceed 100.

The FFC value obtained in Eq. (11) represents soil water uniformly distributed through the top 1.0 m of soil. Runoff estimates can be improved if the depth distribution of soil water is known. For example, a decreasing soil water depth distribution (more water near the soil surface) produces more runoff than a uniform distribution. Conversely, a uniform soil water depth distribution produces more runoff than an increasing soil water depth distribution (more water in lower soil layers). Because continuous simulation models estimate water content of each soil layer daily, the depth distribution is available. The effect of depth distribution on runoff is expressed in the depth weighting function

$$FFC^* = \frac{\sum_{l=1}^M FFC_l \frac{Z_l - Z_{l-1}}{Z_l}}{\sum_{l=1}^M \frac{Z_l - Z_{l-1}}{Z_l}}, \quad Z_l \leq 1.0 \text{ m} \quad FFC' = X1/X2 \quad (16)$$

where  $FFC^*$  = depth weighted FFC value for use in Eq. 12; and  $Z$  = depth in m to the bottom of soil layer  $l$ . Eq. (16) performs two functions: (1) it reduces the influence of lower layers, because  $FFC_l$  is divided by  $Z_l$ , and (2) it gives proper weight to thick layers relative to thin layers because  $FFC_l$  is multiplied by the layer thickness.

Several hydrologic models use the continuous CN direct soil-moisture linkage approach, including CREAMS (Knisel 1980), Groundwater Loading Effects of Agricultural Management Systems (GLEAMS) (Leonard et al. 1987), Simulating Production and Utilization of Range Land (SPUR) (Hansen et al. 1992), Simulator for Water Resources in Rural Basins (SWRRB) (Williams et al. 1985), Water Erosion Prediction Project (WEPS) (Wagner et al. 1995), Erosion Productivity Impact Calculator (EPIC) (Williams et al. 1984), agricultural policy/environmental extender (APEX) (Williams et al. 2000, Williams and Izaurralde 2005a, b),

and the soil and water assessment tool (SWAT) (Arnold et al. 1993). However, for soils with low soil water storage capacity (shallow or coarse textured soils) the direct-linked models tend to over predict runoff because they can be saturated quickly, giving CN values approaching 100.

### The Revised SMI Method

The problems with low soil water storage capacity soils has led to reexamination of the revised SMI approach. The revised SMI method introduced in 1976 (Williams and Laseur) was modified to form the new SMI equations. The  $s$  is driven to between near 0.0 and  $s_1$  or even greater, depending on parameter settings, using the equations

$$s = s_o + PET \exp\left(-P_{42} \frac{s_o}{s_1}\right) - RFV + Q; \quad s < P_{44} \times s_1 \quad (17)$$

where  $s_o$  = previous day's retention parameter; PET = potential evapotranspiration;  $P_{42}$  = SCS CN index coefficient; and  $P_{44}$  = upper limit of the  $s$  parameter. Eq. (17) increases  $s$  rapidly when PET is high and  $s_o$  is low (wet conditions). The revised SMI method has two distinct advantages over the methods that connect  $s$  to soil water directly: (1) the two parameters  $P_{42}$  and  $P_{44}$  are convenient for calibration; and (2) soils with low soil water storage capacity (shallow or coarse textured soils) tend to present major problems for the direct coupled methods because runoff may be over estimated. Kannan et al. (2008) used the SWAT model (Neitsch et al. 2004; Arnold and Fohrer 2005) to test the revised SMI method, considering different PET methods. The EPIC/APEX/SWAT modeling complex offers five options for estimating potential evaporation: Hargreaves and Samani (1985), Penman (1948), Priestley-Taylor (1972), Penman-Monteith (Monteith 1965), and Baier-Robertson (1965). The revised SMI equations are now offered as options in EPIC, APEX, and SWAT.

Although the revised SMI method presented by Hawkins (1978) is more physically based than Eq. (17), it has no provision for calibration. The CN<sub>2</sub> can be adjusted for calibration purposes, but for watersheds with many soil/land-use combinations it is not convenient. Recently, the revised SMI equations were modified to include actual ET and installed in EPIC. The modified equations are

$$s = s_o + ET_{mx} - RFV + Q \quad (18)$$

$$ET_{mx} = \max \left[ ET, \exp \left( -P_{42} \frac{s_o}{s_1} \right) \right] \quad (19)$$

Eqs. (18) and (19) have the convenient calibration parameters ( $P_{42}$  and  $P_{44}$ ) and also the possibility of using actual ET instead of the index. The use of actual ET may be advantageous, especially for winter crops, in which case ET may exceed the index.

Santhi et al. (2001) implemented the revised SMI method in the watershed-scale model, SWAT (version 2005) (Neitsch et al. 2004; Arnold and Fohrer 2005), and applied it for runoff estimation and flow calibration in a large watershed called the Bosque River Watershed in Texas (Fig. 1), which had nearly 30% of shallow soils distributed all over the watershed.

In SWAT, the surface runoff is calculated as a function of rainfall, initial abstraction of rainfall, and soil retention parameter and curve number, as described in Eqs. (1) and (2). The revised SMI method implemented in SWAT allows the retention parameter to vary with accumulated plant evapotranspiration. Calculation of the daily CN value as a function of plant evapotranspiration was added because the soil-moisture method was predicting too much

runoff in shallow soils. By calculating daily CN as a function of plant evapotranspiration, its value is less dependent on soil storage and more dependent on antecedent climate. In the revised SMI method, the retention parameter is updated at the end of every day, as follows:

$$S = S_{\text{prev}} + E_o \times \exp\left(\frac{-\text{cncoef} \times S_{\text{prev}}}{S_{\text{max}}}\right) - R_{\text{day}} + Q_{\text{surf}} \quad (20)$$

where  $S$  = retention parameter for a given day (mm);  $S_{\text{prev}}$  = retention parameter for the previous day (mm);  $E_o$  = potential evapotranspiration for the day ( $\text{mm d}^{-1}$ );  $\text{cncoef}$  = weighting coefficient used to calculate the retention coefficient for daily curve-number calculations dependent on plant evapotranspiration;  $S_{\text{max}}$  = maximum value the retention parameter can achieve on any given day (mm);  $R_{\text{day}}$  = rainfall depth for the day ( $\text{mm H}_2\text{O}$ ); and  $Q_{\text{surf}}$  = surface runoff ( $\text{mm H}_2\text{O}$ ). The initial value of the retention parameter is defined as  $S = 0.9 \times S_{\text{max}}$ . The daily curve-number value [Eq. 2] is adjusted or updated for moisture content with the newly computed retention parameter,  $S$  (Neitsch et al. 2004). The only parameter adjusted for calibrating surface runoff is  $\text{cncoef}$ , instead of adjustment of CN for all combinations of soil and land cover.

## Models Used in Study

### Agricultural Policy/Environmental Extender

The APEX model (Williams et al. 2000; Williams and Izaurralde 2005a, b) was used to compare the two CN methods. APEX was developed for use in whole farm/small watershed management. The history of the APEX model's development was recently reported by Gassman et al. (2010). The model was constructed to evaluate various land-management strategies, considering sustainability, erosion (wind, sheet, and channel), economics, water supply and quality, soil quality, plant competition, weather, and pests. Management capabilities include irrigation, drainage, furrow diking, buffer strips, terraces, waterways, fertilization, forest management, manure management, lagoons, reservoirs, crop rotation and selection, pesticide application, grazing, and tillage. The model operates on a daily time step (some processes are simulated with hourly or less time steps) and is capable of simulating hundreds of years if necessary. Farms may be subdivided into fields, soil types, landscape positions, or any other desirable configuration.

### Soil and Water Assessment Tool

The SWAT (Neitsch et al. 2004; Arnold and Fohrer 2005) is a watershed-scale model developed by the USDA Agriculture

Research Service to predict the impact of land-management practices on water, sediment, and agricultural chemical yields. It is a semidistributed continuous-simulation model that operates on a daily time step. It incorporates simulation of weather, crop growth, evapotranspiration, runoff, soil erosion, and transport of nutrients and pesticides, and divides a watershed into subbasins on the basis of sections of channels. Subbasins, in turn, are divided into hydrological response units, which are unique combinations of soil and land cover. Flow and pollutants from hydrological response units (HRUs) within a subbasin are lumped and routed through the channel. Hundreds or thousands of HRUs can be modeled for a simulation period ranging from one day to several years.

## Applications of Revised SMI CN Method

### Field/Farm-Scale Application Using APEX

Two continuous CN methods were compared using two shallow soils (Purves and Moloterr, 410 and 254 mm in depth, respectively) and two deep soils (Desan and Slidell, both 2030 mm in depth) (Table 1). Desan is very sandy, with very low water storage. These soils, except for Desan, are common in the Bosque River watershed in central Texas. Soil properties are shown in Table 1. A testing area of 65 ha, which is one subarea from a previous APEX model setup for a subwatershed (10,400 ha) within the Bosque River watershed (Wang et al. 2010), was used in this study. The land use for the subarea is Bermuda grass. The upland characteristics and channel properties of the area are shown in Table 2. The APEX model was run for each of the soils for 11 years (1993–2003) using the direct soil-moisture approach (Eqs. 12–15) and the revised SMI method (Eq. 17). The average annual precipitation for the simulation period is 868 mm. The APEX model was previously calibrated and validated during the study period for flow for the revised SMI method in Wang et al. (2010), with a calibrated value of 1.2 for  $P_{42}$  and 1.5 for  $P_{44}$ . These values were used in this study while using the revised SMI method. Simulated results were aggregated at the annual time step for analyzing the performance of direct-link soil moisture-based CN method and the revised SMI CN method.

### Large Watershed-Scale Application Using SWAT

Bosque River Watershed has a drainage area of 4,277  $\text{km}^2$ . Land use in this watershed is mostly range and pasture in the upper portion, whereas cropland is wide spread in the southern portion near the Middle Bosque River. Major soil series are Windthorst (sandy loam, deep, hydrologic group C), Purves (clay, fine, shallow, hydrologic group D), Denton (silty clay, fine, deep, hydrologic group D), Brackett (clay loamy, gravel, deep, hydrologic group C),

**Table 1.** Properties of Soils Used for Testing CN Method

| Soil series | Hydrologic soil group | Initial $\text{CN}_2$ | Depth of layer bottom (mm) | Bulk density ( $\text{ton/m}^3$ ) | Water content (mm) at |                | Sand (%) | Silt (%) |
|-------------|-----------------------|-----------------------|----------------------------|-----------------------------------|-----------------------|----------------|----------|----------|
|             |                       |                       |                            |                                   | Wilting point         | Field capacity |          |          |
| Purves      | D                     | 84                    | 10                         | 1.17                              | 0.290                 | 0.440          | 23.34    | 29.16    |
|             |                       |                       | 410                        | 1.17                              | 0.290                 | 0.440          | 23.34    | 29.46    |
| Moloterr    | D                     | 84                    | 203                        | 1.40                              | 0.188                 | 0.327          | 30.10    | 40.00    |
|             |                       |                       | 254                        | 1.70                              | 0.013                 | 0.102          | 32.30    | 40.00    |
| Desan       | A                     | 49                    | 10                         | 1.75                              | 0.046                 | 0.127          | 86.40    | 6.60     |
|             |                       |                       | 20                         | 1.75                              | 0.046                 | 0.127          | 86.40    | 6.60     |
|             |                       |                       | 137                        | 1.75                              | 0.046                 | 0.127          | 86.40    | 6.60     |
|             |                       |                       | 2,030                      | 1.64                              | 0.109                 | 0.212          | 62.40    | 19.10    |
| Slidell     | D                     | 84                    | 150                        | 1.40                              | 0.280                 | 0.420          | 22.23    | 27.77    |
|             |                       |                       | 810                        | 1.40                              | 0.254                 | 0.420          | 5.26     | 44.74    |
|             |                       |                       | 2,030                      | 1.45                              | 0.247                 | 0.400          | 5.26     | 44.74    |

**Table 2.** Upland and Channel Properties of Study Area

| Area of field (ha) | Land use           | Upland characteristics |                  |                     | Channel properties |                   |                     |
|--------------------|--------------------|------------------------|------------------|---------------------|--------------------|-------------------|---------------------|
|                    |                    | Slope (fraction)       | Slope length (m) | Manning's roughness | Slope (fraction)   | Slope length (km) | Manning's roughness |
| 64.8               | Bermuda grass lawn | 0.05                   | 91.44            | 0.15                | 0.022              | 1.78              | 0.015               |

**Table 3.** Properties of Selected Soils in Bosque River Watershed from SWAT Modeling Using Revised SMI Method

| Soil       | Soil depth (mm) | Deep or shallow soil | Available water holding capacity (mm/year) | Surface runoff (mm/year) | ET (mm/year) | Precipitation (mm/year) | Land use          |
|------------|-----------------|----------------------|--------------------------------------------|--------------------------|--------------|-------------------------|-------------------|
| Windthorst | 1,778           | Deep                 | 223                                        | 85                       | 855          | 968                     | Pasture           |
| Slidell    | 2,632           | Deep                 | 283                                        | 90                       | 848          | 930                     | Sorghum-grain hay |
| Denton     | 1,778           | Deep                 | 266                                        | 127                      | 855          | 986                     | Pasture           |
| Purves     | 508             | Shallow              | 62                                         | 128                      | 717          | 913                     | Range             |
| Aledo      | 508             | Shallow              | 49                                         | 107                      | 659          | 898                     | Range             |
| Eckrant    | 762             | Shallow              | 47                                         | 147                      | 698          | 950                     | Forest            |
| Pidcoke    | 432             | Shallow              | 53                                         | 166                      | 710          | 962                     | Pasture           |

Crawford (silty clay, fine, deep, hydrologic group C), Eckrant (clay, fine, shallow, hydrologic group D), and Aledo (clay loam, gravel, shallow, hydrologic group C). Bosque River Watershed had a considerable area with shallow soils with low soil water storage capacity in addition to deep soils. Distribution of shallow soils above Hico, Texas, were approximately 6%; above Valley Mills they were approximately 22%, and approximately 30% in the entire watershed. The mixed land-use and soil-distribution combinations posed challenges in using the direct-link CN method in runoff estimation and flow calibration in this large watershed. It was necessary to implement the revised SMI CN method in the SWAT model, which was applied for simulating this watershed. The Bosque watershed was discretized into 47 subwatersheds for SWAT modeling. Each subwatershed was then divided into many HRUs, which are unique combinations of soil and land cover. A curve number is assigned to each HRU on the basis of the hydrologic soil group and land cover. Agricultural management practices are assigned to each HRU on the basis of land use. Surface runoff is estimated using the SCS curve-number method for each HRU. Properties of selected soils in the Bosque River Watershed from SWAT modeling using the revised SMI method are presented in Table 3.

Both long-term simulated annual flow and short-term simulated monthly flow from the SWAT model were calibrated with measured flow data (for the time-period data available) at the two most important gauging stations, Hico and Valley Mills on the Bosque River (Fig. 1). Watershed areas above Hico and Valley Mills are 926 and 2,997 km<sup>2</sup>, respectively.

Initially, surface flow and base flow proportions from USGS's daily observed flow data, and simulated daily flow data from SWAT, were calibrated (Fig. 2) for the period from 1960 to 1997 at Hico and Valley Mills, using the automated digital filter technique (Arnold et al. 1995). The annual average water yield of this watershed is 150 mm, of which 30% is in the form of baseflow (subsurface runoff); the rest is in the form of surface runoff. The model was calibrated to total annual stream flow from 1960 to 1997. For flow calibration, the runoff curve numbers (CN2) were adjusted within 10% from the tabulated curve numbers (Mockus 1969) after the implementation of the revised SMI method to reflect conservation tillage practices and soil residue cover conditions of the watershed. Other flow-related model parameters, such as the reevaporation coefficient for groundwater (REVAPC), which represents the water that moves from the shallow aquifer back to the soil profile/root zone and plant uptake from deep roots (Arnold et al. 1993), soil evaporation compensation factor (ESCO), and plant

evaporation compensation factor (EPCO), were adjusted from SWAT initial estimates to match the simulated and observed flows (Table 4). The model was then run from 1990 to 1998 for validation of monthly flow. Model calibration parameters remained the same for the long-term annual flow calibration and the short term monthly flow calibration.

Mean, standard deviation, coefficient of determination ( $R^2$ ), and Nash-Sutcliffe simulation efficiency ( $E_{NS}$ ) (Nash and Sutcliffe 1970) were used to evaluate model prediction. The  $R^2$  value is an indicator of the strength of relationship between the observed and simulated values. If the  $R^2$  value is less than or very close to zero, the model prediction is considered unacceptable or poor. If the value is one, then the model prediction is perfect. If the values are greater than 0.5, then the model prediction is good. Nash-Sutcliffe simulation efficiency ( $E_{NS}$ ), or explained variance, indicates both correlation of observed and simulated values and agreement of the means. The Nash and Sutcliffe efficiency is computed as follows:

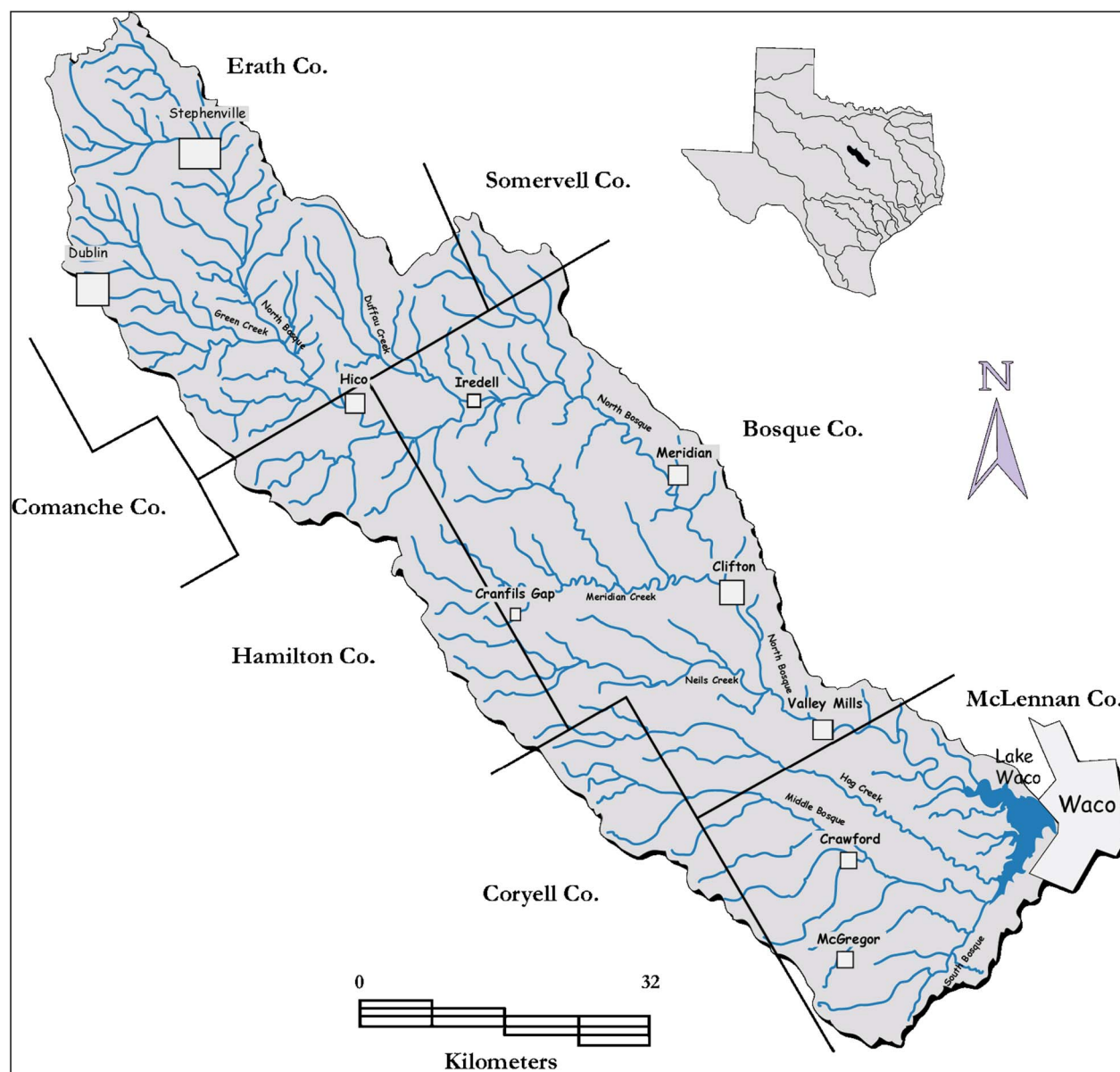
$$E_{NS} = 1 - \frac{\sum_{i=1}^n (o_i - p_i)^2}{\sum_{i=1}^n (o_i - \bar{o})^2} \quad (21)$$

where  $E_{NS}$  = Nash and Sutcliffe efficiency;  $O$  represents observations of flow;  $\bar{O}$  = mean of flow observations;  $P$  represents modeled flow results; and  $i$  = index (varies from 1 to total number of observations).

$E_{NS}$  is the ratio of variance in predicted and observed values to the variance in observed values (with respect to the observed mean). The efficiency can range from  $-\infty$  to 1. A value of 1 represents a perfect model fit between predictions and observations, and values less than 0 indicate that the mean of observations would have been a better estimator for prediction of flow values than the model considered. In general, an  $E_{NS}$  value of 0.5 and above is acceptable.

### River Basin—Scale Application Using SWAT

An ongoing national assessment study called the Conservation Effects Assessment Project (CEAP) uses the Hydrologic Unit Modeling for the United States (HUMUS) framework (Srinivasan et al. 1998), comprising the SWAT model with updated databases for the 18 major water-resource regions. HUMUS was designed for making assessments of national and river basin—scale water demands and land-management practices. The study uses the revised SMI method for modeling flow of all river basins. Results are



**Fig. 1.** Bosque River Watershed in Texas [Reprinted with permission from Santhi et al. (2001)]

analyzed at the 8-digit watershed level. Some specific examples from the CEAP study for two 8-digit watersheds are presented here in the context of revised SMI CN method. The characteristics of the study are presented in Table 5. The two 8-digit watersheds in the study area had a large amount of shallow soils (Table 5), and therefore the revised SMI method was used for estimating runoff. The modeling framework used Penman-Montieth method for estimation of evapotranspiration. Modeling was carried out at annual time step from 1961 to 2006. Thirty years of data from 1961 were used for calibration and the remaining for validation. More details on data used, modeling framework, sensitivity of model parameters, and calibration details are available in Kannan et al. 2008.

## Results and Discussion

### Small-Scale Application Using APEX

Simulated results from APEX model runs using the two CN methods are presented in Table 6. Surface runoff and ET were considered for the evaluation of the CN methods. Purves and

Moloterr are shallow soils and Slidell is deep soil, and all classified as hydrologic group D (initial  $CN_2 = 84$ ). Although Desan is as deep as Slidell, Desan is very sandy (>80% with low storage, and is classified as hydrologic soil group A (initial  $CN_2 = 49$ ). Therefore, surface runoff from the Desan soil is much less than from the other three soils (Table 6). For shallow soils or soils with low water storage the simulated surface runoff by the direct soil-moisture approach is higher than that by the revised SMI method [Fig. 3(a)], although the ET values are almost equal (Table 6). However, for deep soils with water storage capacities of  $0.13 \text{ m m}^{-1}$  or greater, both methods give similar results. For example, the two CN methods predicted approximately the same annual surface runoff for the Slidell soil [Table 6; Fig. 3(b)]. The average annual runoff for the Bosque River watershed at the stream gauge near Valley Mills, Texas, is approximately 85 mm. Approximately 35% of the soils in the watershed are shallow. Using the direct-link method produces an average annual runoff of approximately 250 mm from the shallow soils. This implies that the remaining 65% of the watershed produces negative runoff. Therefore, if the direct-link method is used in modeling, the initial  $CN_2$  values for shallow soils

or soils with low storage have to be manually adjusted (reduced) to overcome the overprediction. Adjusting CN values for certain soils in a watershed is arbitrary, not convenient, and more difficult for automatic calibration procedures.

Also, by examining the simulated CN distributions (Fig. 4) for the two methods, it is clear that the direct-link method produces more extreme runoff events (both high and low) for shallow soils or soils with low storage. For the Moloterr soil, 51% of the simulated storms had CN values greater than 90, and 3% were below 60 using the direct-link method. In contrast, 9% of the simulated CN values were above 90, and 2% were below 60 using the revised SMI method [Fig. 4(a)]. However, both methods produced very similar CN distributions for the deep Slidell soil [Fig. 4(b)], which has  $0.16 \text{ m m}^{-1}$  water storage capacity (Table 1). The obvious problem with using the direct-link method to simulate runoff from shallow soils is small storage capacity that fluctuates between wet and dry much too rapidly. This indicates that the revised SMI method is more robust and that for shallow soils or soils with low storage, the revised SMI method is more appropriate than the direct-link method.

### Watershed-Scale Application Using SWAT

Measured and simulated annual flows at Hico and Valley Mills matched well [Figs. 2(a and b)]. Monthly simulated and observed

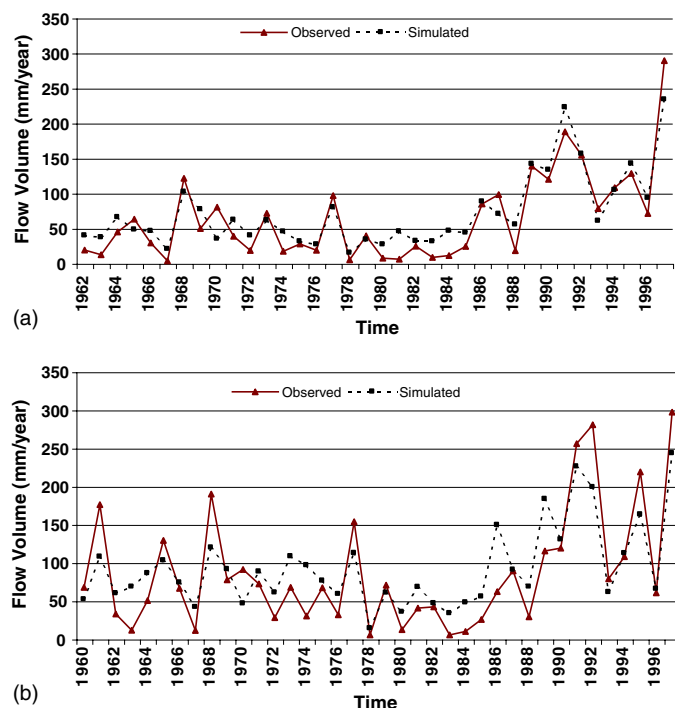


Fig. 2. Observed and simulated annual flows: (a) Hico; (b) Valley Mills

Table 4. Model Parameters Adjusted in SWAT for Calibration of Flow in Bosque River Watershed

| Parameter                           | Description                                                                                                                                           | Range     | Calibrated value of parameter |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------|
| CN <sub>2</sub>                     | Curve number                                                                                                                                          | +10%      | +10%                          |
| Ground Water ReEvAPoration (GWREVP) | Groundwater reevaporation coefficient: it controls the upward movement of water from shallow aquifer to root zone in proportion to evaporative demand | 0.00–1.00 | 0.03                          |
| ESCO                                | Soil evaporation compensation factor: it is used to modify the depth distribution of water in soil layers to meet the soil evaporative demand         | 0.00–1.00 | 0.60                          |
| EPCO                                | Plant evaporation compensation factor: it allows water from lower soil layers to meet the potential water uptake in upper soil layers                 | 0.00–1.00 | 0.10                          |

flows during calibration at the two locations matched well. Means and standard deviations of the observed and simulated flows for long term and short term at Hico and Valley Mills were within a difference of 10% (Table 7). Further agreements between observed and simulated flows at Hico and Valley Mills are shown by the coefficient of determinations greater than 0.6 and the  $E_{NS}$  value greater than 0.72 (Table 7). The revised SMI CN method was very useful for flow calibration at various gauges in this large watershed with mixed land uses and deep and shallow soils.

### River Basin—Scale Application Using HUMUS-SWAT

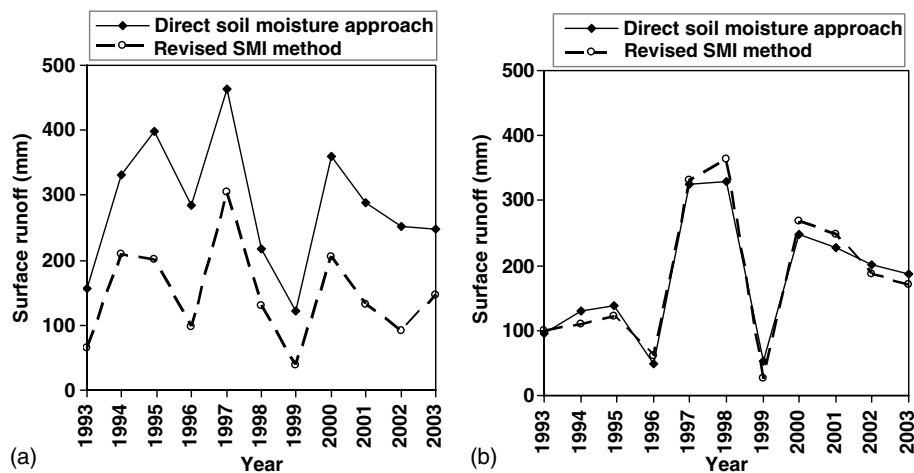
Selected results from the CEAP study for two 8-digit watershed are shown in Table 8. The table shows annual predicted results of surface runoff, subsurface flow, and water yield, along with observations using the soil moisture—based and revised SMI CN methods. For making additional assessments on results, annual precipitation and ET are also shown. Both the 8-digit watersheds contain shallow soils for more than 90% of the total area. However, the amount and pattern of precipitation is different. The table shows that the different CN methods are consistent in estimating ET. However, the estimated surface runoff and water-yield values are different. The estimated results are closer to the observations when using the revised SMI CN method (Table 8). The advantage of the revised SMI method in estimating runoff is apparent from Table 8. The effect is prominent, probably because the precipitation is less (8-digit watershed 12090109). In this case, the soil moisture—based CN method is estimating more runoff because the soils are at or more than field capacity more often. However, the revised SMI method predicts a reasonable amount of surface runoff and

Table 5. Characteristics of Study Area for River Basin—Scale Application of Revised SMI CN Method

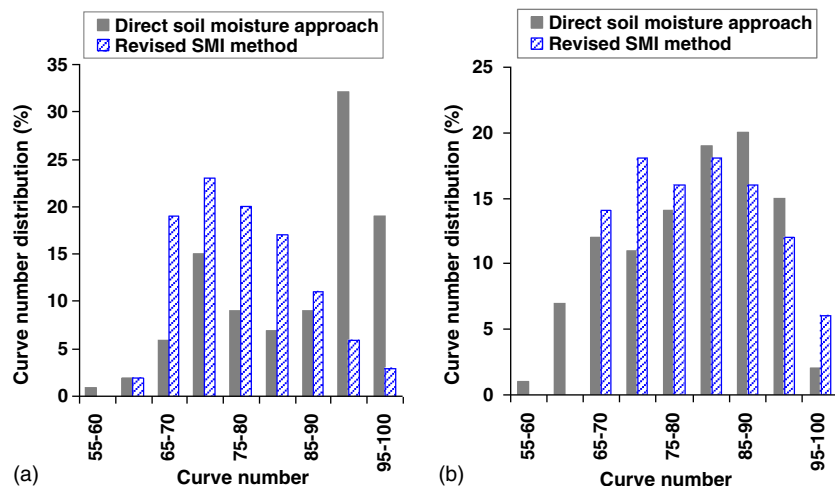
| US water resource region | 8-digit watershed | Mean depth of soil (mm) | Proportion of watershed under shallow soils |
|--------------------------|-------------------|-------------------------|---------------------------------------------|
| New England              | 01030002          | 460                     | 93                                          |
| Texas Gulf               | 12090109          | 380–760                 | 95                                          |

Table 6. Predicted Surface Runoff by Different CN Methods

| Soil series | Direct-link soil-moisture-based CN method |                     | Revised soil-moisture index method |                     |
|-------------|-------------------------------------------|---------------------|------------------------------------|---------------------|
|             | ET (mm)                                   | Surface runoff (mm) | ET (mm)                            | Surface runoff (mm) |
| Purves      | 570.2                                     | 165.6               | 568.5                              | 118.3               |
| Moloterr    | 409.1                                     | 252.2               | 419.9                              | 128.8               |
| Desan       | 608.9                                     | 5.3                 | 608.9                              | 0.4                 |
| Slidell     | 696.2                                     | 182.7               | 694.0                              | 180.6               |



**Fig. 3.** Simulated annual surface runoff using the direct soil-moisture approach and the revised SMI method for shallow soil and deep soil: (a) Moloterr soil, depth of 254 mm; (b) Slidell soil, depth of 2,030 mm



**Fig. 4.** Curve-number distributions during the simulation period (1993–2003) by direct soil-moisture approach and revised SMI method for shallow soil and deep soil, both in hydrologic soil group D: (a) Moloterr soil, depth of 254 mm; (b) Slidell soil, depth of 2,030 mm

**Table 7.** Summary Statistics for Annual and Monthly Flow Calibration at Hico and Valley Mills in Bosque River Watershed in Texas

| Variable (units)                 | Mean     |           | Standard deviation |           | $R^2$ | $E_{NS}$ | Calibration period |
|----------------------------------|----------|-----------|--------------------|-----------|-------|----------|--------------------|
|                                  | Observed | Simulated | Observed           | Simulated |       |          |                    |
| Annual flow at Hico, TX (mm)     | 64.46    | 72.09     | 62.28              | 52.97     | 0.88  | 0.86     | 1962–1997          |
| Annual flow at Valley Mills, TX  | 86.50    | 92.31     | 77.05              | 53.23     | 0.66  | 0.72     | 1960–1997          |
| Monthly flow at Hico, TX         | 12.56    | 11.99     | 15.07              | 14.55     | 0.81  | 0.80     | 1993 Jan–1998 Jul  |
| Monthly flow at Valley Mills, TX | 15.29    | 14.24     | 24.95              | 16.18     | 0.87  | 0.79     | 1996 Jan–1998 Jul  |

**Table 8.** Calibrated Average Annual Results with Different CN Methods from River Basin Scale–Application Studies

| Observations/Modeled results | 8-digit watershed | Mean annual values (mm) of |       |                |                 |             |
|------------------------------|-------------------|----------------------------|-------|----------------|-----------------|-------------|
|                              |                   | Precipitation              | ET    | Surface runoff | Subsurface flow | Water yield |
| Observations/target          |                   |                            |       | 291.7          | 331.1           | 622.9       |
| Soil moisture CN method      | 01030002          | 1051.5                     | 476.1 | 315.7          | 261.7           | 577.4       |
| Revised SMI CN method        |                   |                            | 475.2 | 312.0          | 267.7           | 579.7       |
| Observations/target          |                   |                            |       | 8.5            | 5.7             | 14.2        |
| Soil moisture CN method      | 12090109          | 613.9                      | 494.6 | 53.9           | 6.1             | 60.0        |
| Revised SMI CN method        |                   |                            | 497.1 | 8.1            | 5.9             | 13.9        |

water yield because it is based on the value of water depletion or ET. A similar effect is occurring for the 8-digit watershed 01030002. However, the effect is not very prominently seen from the results. High total rainfall (resulting from frequent rain events) and frequently saturated soils are probably masking some of the effects of the revised SMI method.

## Summary and Conclusions

The SCS used the CN equation in a continuous mode to evaluate floodwater-retarding projects in the early years (1950–1980). Runoff was estimated from daily rainfall records using the five-day antecedent rainfall to assign a  $CN_1$  (dry condition),  $CN_2$  (average condition), or  $CN_3$  (wet condition) to each storm. The continuous CN approach was modified (Williams and LaSeur 1976) by replacing the five-day antecedent rainfall with an SMI. The revised SMI approach formed the basis for the use of the continuous CN method in several watershed simulation models. However, it was generally thought that a direct link between the SCS retention parameter  $s$  and soil water content would be more physically correct than the revised SMI. Several direct-link equations were developed and installed in the hydrology components of models including CREAMS, GLEAMS, SPUR, SWRRB, WEPS, EPIC, APEX, and SWAT. As the models were applied, it became apparent that there was a problem in simulating runoff from soils with low soil water storage capacity. The direct-linked models tend to overpredict runoff because they can be saturated quickly. This problem led to a reexamination of the revised SMI approach. The revised SMI method overcame the problem with low storage soils and provided a convenient way to calibrate.

*Publisher's Note. This paper was posted ahead of print with an incorrect author list. The correct author list appears in this version of the paper.*

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